THE TRADING DESK THAT SEES CONTAGION BEFORE PRICES DO

Knowledge Graphs, Obsidian Vaults, Evidence Retrieval, and Bounded AI Judgment

CENTRAL PROPOSITION

A graph-grounded Second Brain can turn a fragmented institutional knowledge base into traceable evidence about complex, indirect economic exposures. Deterministic algorithms construct and evaluate possible transmission paths; a language model interprets the evidence within bounded authority; human judgment governs any resulting decision.

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Second Brain Laboratory Series | Lab 2 | Pedagogical Research Monograph

Entire laboratory is synthetic and educational. No company, shock, edge, market forecast or recommendation is real.

Abstract

This monograph presents a ten-stage pedagogical laboratory for building and evaluating an institutional Second Brain for synthetic market intelligence. It asks whether typed relationships among firms can make indirect dependencies easier to discover and audit after a hypothetical semiconductor production disruption. The setting contains five hundred fictional companies, documented as a registry, a relationship ledger, and individual Markdown dossiers. Every entity, edge, financial attribute, event, and score is invented for instruction; none represents an actual issuer, contract, market event, or investment view.

The workflow separates durable evidence from interpretation. It establishes a reproducible environment, creates the synthetic corporate universe, records typed relationships and provenance, validates a directed multigraph, and propagates a bounded operational shock along supply links. It then constructs event-only and broad-document baselines, compares them with transparent TF–IDF retrieval, and prepares a graph-aware package containing ranked firms, transmission paths, and source identifiers. These operations are deterministic under the stated inputs and assumptions. Financing and ownership links remain visible as distinct relationship classes but are not treated as interchangeable physical transmission routes.

An optional Claude Haiku 5.5 call receives only the prepared graph-aware evidence and is asked to produce a cautious, reviewable market-risk memorandum. It does not create the underlying network, calculate the exposure model, fetch prices, or place trades. When the API is unavailable or returns an error, the notebook preserves that status instead of inventing a response. Human reviewers remain responsible for evaluating source quality, assumptions, uncertainty, and any proposed safeguards. Readers should approach the resulting memorandum as a review artifact, not as an answer to the market question. Its value depends on whether each claim can be traced to an input record, whether the propagation assumptions match the scenario, and whether uncertainty remains visible when evidence is incomplete. The exported archive supports that review by preserving intermediate files rather than only a polished conclusion. Instructors can then ask learners to modify one assumption, rerun the stages, compare the evidence packages, and explain why a finding changed. Such controlled experiments teach where a knowledge graph helps, where it fails, and which validation is still required. A future empirical study would require independently sourced company relationships, time stamped outcome data, out of sample evaluation, and a separately specified standard for useful human decisions. Until then, results should be treated as demonstrations of traceable workflow design, not financial advice or evidence of predictive advantage when used by practitioners.

RESEARCH QUESTION

Can an institutional knowledge graph reveal traceable, indirect dependencies after a synthetic semiconductor disruption and support a more evidence-grounded market-risk memorandum than an unordered collection of notes?

COMPUTATIONAL COMPANION

The authoritative technical companion is `TRADING_CONTAGION_SECOND_BRAIN.ipynb`, included in this bundle. Its ten numbered laboratory stages are mapped one-to-one to the ten infographic files `cell_1.png` through `cell_10.png`. The notebook and this monograph form the reference record for the claims discussed here.

Introduction: Institutional knowledge as a risk instrument

— Institutional knowledge meets market intelligence

Financial markets are often analyzed by looking first at market prices. Returns, volatilities, correlations and estimated factors provide an indispensable language for risk. Yet a trading desk confronted with a new disturbance faces a different question: which firms are structurally exposed before the historical return series has enough observations to register the new transmission mechanism? A factory outage can interrupt an input, which delays a component, which constrains an assembler, which weakens a borrower, which affects a lender. These dependencies are economic facts and organizational knowledge before they become statistical regularities.

This laboratory makes that distinction operational. The universe contains five hundred synthetic companies and multiple networks: supplier–buyer dependencies, common financing channels, equity ownership, sector affinity, and geographic concentration. Each relationship is given a type and a direction. The result is not merely a graph of who knows whom. It is a record of how disturbances may travel and of why a particular edge should matter for a specified event. Although the data are invented, the analytical challenge resembles the work of investment committees, risk offices, supply-chain analysts, and financial supervisors.

The starting point is the Second Brain conception developed in Laboratory 01. External institutional memory stores durable notes. Metadata preserves identifiers and provenance. A typed knowledge graph captures dependencies. Deterministic tools calculate and constrain. A large language model receives selected evidence, distinguishes what is known from what is conjectured, and prepares a compact, reviewable recommendation. Governance and export make the chain reproducible. The move from regulation to market intelligence is deliberate: it tests whether the same architecture generalizes across a domain boundary while respecting the specific semantics of financial dependencies.

The case begins with a hypothetical disruption at selected semiconductor manufacturers in East Asia. We introduce direct production impairment and let its effects pass through supplier–customer edges with diminishing intensity. We separately consider shared lenders and ownership networks as channels for second-round vulnerability. Not every network connection is causally equivalent. A supply edge may transmit operational scarcity; common ownership may transmit portfolio-level concentration but not a physical production stoppage. The program therefore records edge types and explicitly weights them, instead of collapsing every link into a generic relevance score.

The laboratory teaches an important form of disciplined imagination. Traditional covariance matrices summarize co-movement during the estimation window. They are useful for portfolio construction, but they do not encode all possible future operational linkages. A company can be a major indirect customer of a disrupted input even if its historical returns were weakly correlated with the initial supplier. Conversely, a high-correlation peer might have no meaningful direct dependence on the affected component. Our objective is not to demonstrate that correlations are useless; it is to isolate their informational limits during a genuinely new structural shock.

The second major teaching point concerns the difference between retrieval and navigation. A lexical or TF–IDF search can locate notes mentioning semiconductors. An event-conditioned graph traversal can follow a supplier into an assembler and then into a financed distributor, assembling a chain of evidence even if the final firm's note never contains the term semiconductor. This is the knowledge advantage of relational context. It is also a liability when edges are wrong, stale, weakly substantiated, or excessively permissive. We therefore retain edge provenance, maintain bounded traversal, and make graph-generated claims inspectable.

The exercise intentionally separates three operations often confused in discussions of AI. First, an economic model computes deterministic exposure indicators under transparent simplifying assumptions. Second, an information architecture selects and packages the evidence used to discuss those indicators. Third, Claude Haiku 5.5, when authorized and available, proposes judgment under explicit restrictions. The model does not manufacture market observations, run trades, or silently rewrite the graph. Its task is to articulate plausible risk responses, limitations and monitoring triggers from the context supplied by the preceding cells.

Four experimental conditions support comparison. An event-only baseline knows the shock narrative but not the dependency structure. A broad-document context concatenates a bounded sample of company notes. A TF–IDF context retrieves firms using word overlap. A graph-aware context follows typed dependencies and preserves the exposure calculation with source identifiers. Because the contexts carry different information and lengths, the experiment is an architectural demonstration rather than a controlled estimate of causal performance. We make this caveat explicit when interpreting recall, evidence efficiency, and any generated memorandum.

The synthetic portfolio includes representative positions, allowing us to connect knowledge retrieval to a practical risk-management question. A sensible output does not need to say "buy" or "sell." It may recommend deeper

counterparty review, temporary concentration limits, scenario hedges, or additional data collection. A genuine institutional decision also requires liquidity, execution costs, mandates, compliance, and human authorization, none of which this simplified exercise fully models. The notebook therefore yields a decision-support memorandum, not autonomous trading instructions.

Finally, the laboratory addresses institutional memory as a process rather than a static archive. The system exports Markdown notes, the relationship ledger, the event definition, the propagation results, retrieved context packages, evaluation tables and the optional model response. Each item can be reloaded and challenged. Students can change the disrupted manufacturers, weaken a transmission parameter, delete an edge, or add a financing concentration, then compare how the evidence and recommendation change. These controlled counterfactuals are the bridge from passive corporate knowledge to intelligence capable of supporting accountable action.

The essential lesson is that prices are not the only data in finance. Relationships, contracts, physical dependencies and institutional experience form an additional informational layer. When that layer is governed, searchable and connected to bounded reasoning, it can create earlier and more explainable hypotheses about vulnerability. Whether those hypotheses are correct remains an empirical question. A Second Brain improves the discipline with w

THE CENTRAL ARCHITECTURAL DISTINCTION

The system is not an autonomous trading agent. Its evidence is created and linked in the vault; graph algorithms select and propagate relationships; information retrieval assembles relevant context; Claude provides bounded analytical synthesis only if an API response is returned; humans retain responsibility for challenging assumptions and deciding whether any safeguards are warranted.

Cell 01 — Environment, reproducibility, and secrets

We begin by fixing the computational conditions under which the experiment can be repeated. A trading desk cannot audit a result when random draws, package versions or model parameters are implicit. We therefore seed the synthetic data generator and create an explicit project directory. The secret used to access Claude is optional: all purely deterministic stages must work without a paid API or model access. This preserves a clean boundary between evidence construction and the judgment engine.

A second consideration is the difference between demonstrative and deployable code. The notebook is deliberately accessible to financial practitioners who may not write software professionally. However, accessibility must not mean hiding dependencies or swallowing errors. The setup cell checks its environment, installs only absent libraries, and records whether a model key was found without printing its value. No external market feeds are contacted and no actual order submission capability exists.

The resulting working directory functions as an evidence workspace. Later cells will write a Markdown vault and serialized tables into it; the final cell will make a reproducible archive. In a live institutional setting this workspace would be subject to identity controls, information classification, retention rules and quality review. For our teaching example, transparent local files are a better starting point than an opaque service. The central lesson is that an institutional Second Brain begins with disciplined recordkeeping, not with a chatbot interface.

Reproducibility has several layers here. A fixed random seed makes the synthetic population and its relationships repeatable within the notebook's chosen software environment; it does not make those draws economically representative. The directory convention makes generated artifacts easy to locate, while explicit checks help reveal whether expected files were actually written. Students should distinguish a rerunnable cell from a fully reproducible research workflow: the latter also requires recording dependency versions, preserving source inputs, documenting changes, and explaining which outputs depend on optional external services. The laboratory introduces these habits without claiming to provide a production deployment blueprint.

Credential handling is equally important because an API key is a capability, not ordinary research data. The notebook checks whether a key is available through the designated secret store, but the key itself should never appear in printed output, Markdown, screenshots, or exported archives. A missing credential should disable only the model call, not the registry, graph, stress calculation, or retrieval comparisons. That fail-independent design makes it possible to teach the full evidence pipeline without requiring every learner to send data to a provider. It also reduces the temptation to confuse successful setup with successful analysis.

Before moving on, learners can deliberately test the boundaries: rerun with the same seed, inspect the output directory, remove the optional key, and confirm that deterministic artifacts remain available. They should also record environment changes rather than silently updating packages mid-experiment. In a regulated institution, the corresponding controls would include approved dependencies, access logs, retention schedules, and an owner for each data product. Our small local workspace is only a teaching analogue, but the discipline is transferable: control where evidence is written, protect capabilities, and make failures visible.

Teaching perspective: foundations before intelligence

The working directory is a controlled evidence workspace. Its value is not the directory name itself but the principle that every subsequent artifact has a stable location and can be inspected independently of the conversational model. Credentials are separated from the records. This is the first governance boundary: a language model may contribute interpretation later, but the creation, storage and availability of evidence must not depend on it. In this laboratory the Google Drive mount is a convenient instructional medium, not a production-grade database. Hundreds of small Markdown files may expose the limitations of network-mounted storage; retry and verification logic therefore belong to the operational design.

What learners should verify

Check the seeded random generator, the active `ROOT` path, the presence of the optional API key without disclosing it, and the distinction between a successful environment setup and a successful model invocation.

Cell 02 — Build the 500-company institutional universe

The second stage creates a controlled universe large enough to exhibit nontrivial network behavior but small enough to inspect. Each company receives a stable identifier, a sector, a region, a descriptive business activity and simple synthetic financial indicators. These attributes are not forecasts or observations. They are pedagogical scaffolding that lets us ask which exposures are direct, which are indirect and which arise from concentration rather than operational dependency.

The corporate population includes manufacturers of semiconductors, components, devices and industrial products as well as distributors, transport firms, retailers, banks and software companies. Its heterogeneity is important. A purely homogeneous network might make graph traversal look impressive while offering few meaningful distinctions. Here a disruption in semiconductor output should have a stronger immediate impact on companies sourcing chips than on a financial intermediary with no direct operating dependence. The latter might nevertheless be exposed through lending or investment positions.

To emphasize reproducibility, the cell constructs the population using a fixed seed. The variables are intuitive rather than optimized: simplified revenue, leverage and cash-buffer proxies make the subsequent risk score easy to teach. These values should not be mistaken for actual firm balance sheets or a calibrated contagion model. We reserve definitive empirical claims for a future version with authenticated issuer, supply-chain and market data. The outcome of this cell is a machine-readable corporate registry, which becomes the first layer of institutional memory.

The registry is designed to support questions, not to answer them by itself. Stable identifiers allow a firm to be followed consistently from a table row to a graph vertex, a Markdown dossier, a retrieved candidate, and an exported evidence path. Sector and region labels make it possible to define the shock seeds transparently and to inspect whether later results are concentrated in particular parts of the synthetic universe. The descriptive business activity gives text retrieval something to search, while the numerical proxies support a simple illustration of how balance-sheet characteristics can modify an operational exposure score.

The population's size is a deliberate compromise. Five hundred entities provide enough potential routes for multi-hop discovery and enough irrelevant records to make retrieval meaningful, but they remain small enough for students to inspect and for the notebook to run interactively. The ten-sector and five-region structure creates diversity without implying a carefully sampled representation of global industry. The seed fixes one reproducible realization; it does not remove model uncertainty or demonstrate that the chosen proportions match any real economy. A learner changing the seed may create a different network and should expect downstream rankings to change with it.

The cell therefore establishes a controlled experimental substrate. It does not establish which companies are truly systemic, which balance sheets are fragile, or what price response a shock would produce. Its variables are normalized teaching devices whose role is to make later operations legible. An analyst should keep the registry's data dictionary close at hand, identify units and missing-value conventions, and avoid reading synthetic values as precise economic measurements. In a real extension, each field would need a source, an as-of date, a definition, and a documented transformation before it could responsibly inform risk review.

What the registry represents

The registry defines 500 fictional firms spanning ten sectors and five regions, with stable IDs, sector and geographic labels, a synthetic revenue measure, leverage and a cash-buffer variable. It makes the subsequent network exercise testable: each vertex is tied to an identifiable synthetic entity, not an anonymous dot on a chart. A company record alone, however, cannot reveal whether the company supplies a particular automobile manufacturer or borrows from a bank. Those facts belong to the relationship architecture developed next.

Teaching distinction

A *company register* describes entities. A *knowledge graph* describes entities *and* their typed relationships. A *Second Brain* connects these records to durable, inspectable evidence and makes them usable for retrieval and deliberation.

Cell 03 — Relationship records and an Obsidian vault

A list of companies is not yet a Second Brain. Intelligence begins when the records capture how companies depend on one another. This stage creates several directed and typed link families. The direction of a supply edge runs from supplier to customer, matching the direction in which a shortage might propagate. A financing edge runs from lender to borrower and signals a possible credit exposure. An ownership edge records a synthetic investor-to-investee relation and is deliberately treated differently from the physical supply network.

Each link receives a source ID, confidence and effective-date interval. These features are modest stand-ins for a richer provenance system. They let a student inspect why a connection exists, whether it is current for the simulated event date and whether it should be regarded as certain. The vault also receives one Markdown note per company, with YAML-style frontmatter and Obsidian-compatible wikilinks. The notes are readable without Python, which matters if analysts or compliance teams later need to challenge the source material.

We intentionally generate the link ledger and the notes from one common canonical record, reducing the chance that the display and the graph disagree. In a real environment, one would additionally require source-specific permissions, version snapshots, supplier attestations and procedures for managing disputes. For the present laboratory, the most important discipline is simple: economic claims must point to durable records rather than appearing as untraceable statements in a model response.

The canonical ledger is the bridge between machine processing and human review. A source identifier attached to a relationship makes it possible to ask what generated the edge, while confidence and effective dates make uncertainty and timing explicit rather than burying them in prose. The values are synthetic in this laboratory, so provenance fields demonstrate a schema rather than establish independent verification. A high confidence score here means only that the simulation assigned a value; it is not equivalent to a supplier confirmation or a signed contract. Learners should preserve that distinction whenever they summarize paths later.

The Obsidian vault is generated from the same underlying company and relationship records as the ledger. This common origin reduces accidental divergence, such as a note naming a supplier that is absent from the graph or an edge referring to an unknown company. Wikilinks make movement among counterparties natural for a human reader, while frontmatter gives software a predictable place to find IDs and attributes. A person can browse the notes without running code, and a program can continue to use the tabular ledger without scraping rendered text. Neither representation is sufficient alone: one supports narrative inspection and the other supports structured computation.

This design also makes update governance visible. In a live vault, relationships can change; a supplier may be replaced, a lending facility may mature, or an ownership stake may be sold. The effective-date interval should therefore be checked against the scenario date, and superseded records should not simply disappear if an audit needs to explain an earlier decision. Access control, version history, source licensing, and correction procedures would be necessary before institutional use. The laboratory cannot solve those governance problems with Markdown, but it makes them concrete by giving each relationship a durable identifier and a place in the evidence chain.

Obsidian as an institutional evidence interface

The Markdown dossier is a human-readable surface over the canonical ledger. Each company has an Obsidian-compatible note with a consistent identity, financial attributes, inbound and outbound supplier, lender and ownership relationships, and wikilinks to counterparties. Evidence identifiers belong to edges, so a proposed connection can be traced back to a source record rather than merely asserted in prose. The vault does not replace the machine-readable relationship ledger: it complements it. The ledger provides computational consistency, while the vault makes the data inspectable and navigable for non-programmers.

Important limitation

Synthetic evidence IDs identify simulation records, not independently authenticated contracts. Obsidian links make a relationship navigable; they do not independently prove its economic significance.

Cell 04 — Construct and validate the typed corporate graph

The fourth stage turns corporate memory into a computational structure. Nodes represent firms, and edges represent documented relationships. Because the three link families have distinct economic meanings, we use a directed multigraph rather than flattening all relationships into one undifferentiated network. This design preserves parallel links: a supplier can also be owned by or financed by another firm.

Validation must precede analysis. The code checks that every link endpoint belongs to the registry, that edge identifiers are unique, and that all weights and confidence values fall within defined bounds. It then summarizes the number of nodes and edges, the distribution of link types and the largest connected portions of the supply network. These checks are not merely programmer conveniences; they prevent analytical stories from being built on malformed evidence.

The cell also creates an illustrative graph view of a small subnetwork. It is not necessary to draw all five hundred nodes to understand the method. A whole-universe hairball can conceal more than it reveals. The substantive object is the typed and inspectable graph in memory, supported by the CSV ledger. Later stages will use it to identify pathways and assemble context. Students should distinguish topology, causal interpretation and legal or organizational authority: a link states a synthetic relationship, not proof that a shock will transmit with precisely the chosen coefficient.

Validation is treated as an analytical control because graph errors can produce plausible-looking but false pathways. A missing endpoint could silently detach an exposure; a duplicated edge identifier could make a source citation ambiguous; and an out-of-range weight could overwhelm the intended decay. Confirming node membership, identifier uniqueness, and allowed ranges catches these problems before propagation. Summary counts and type distributions provide a second, high-level check: unexpectedly few supply links or an unusual concentration of one relation class can signal that generation or parsing did not behave as expected. These summaries are diagnostics, not evidence that the synthetic network resembles a real economy.

The multigraph preserves parallel relationships instead of collapsing them. Two firms may have a supply connection and a financing connection at the same time, and those edges answer different questions. If all links were reduced to a single adjacency matrix, the analyst would lose the mechanism, direction, and provenance needed to interpret the route. The separate supply-only graph makes the operational stress model explicit and prevents a financing edge from being treated as if it transported a missing component. Other channel-specific models could be added later, but each would require its own assumptions, parameters, and validation.

Visualization is useful mainly as a diagnostic interface. A readable subnetwork can expose direction mistakes, isolated nodes, or unintuitive paths that a table obscures. Conversely, node placement in a drawing is not evidence of economic centrality, and a visually prominent cluster does not establish causal importance. Students should move between graph view and source ledger, verify representative edges, and check that each route corresponds to the intended edge type. This sequence models an important professional habit: use visualizations to formulate questions, then answer them with inspectable records and explicit rules.

Why two network objects are used

The full `MultiDiGraph` retains simultaneous relationship types between counterparties. A separate directed `SUP` network contains only the `supplies` links, oriented from upstream firm to downstream buyer. Restricting propagation to the supply layer is a modeling decision, not a claim that finance and ownership are irrelevant. It prevents the laboratory from applying the same physical-shortage equation to fundamentally different channels.

Graph semantics

Vertex: company ID, sector and region. **Typed edge:** source, target, relationship class, weight, confidence and evidence ID. **Traversal:** following directed links subject to explicit economic rules.

Cell 05 — Semiconductor shock and stress propagation

The fifth stage introduces the event that turns a stored knowledge base into a decision-support system. We select several semiconductor manufacturers located in the synthetic East Asian segment as initially disrupted firms. The selection is deliberately explicit: the shock should not emerge from a hidden rule, because analysts must be able to review whether a scenario captures the intended event.

We then propagate operational disruption along supplier-to-customer edges. The calculation starts with direct severity at affected manufacturers. At each step, a fraction of a supplier's disruption reaches its customers, weighted by a synthetic dependency coefficient. Contributions are bounded and decayed so that the result remains interpretable over a small number of hops. We record the path and supporting evidence IDs whenever a downstream firm is reached. This is a simplified stress-transmission exercise, not a causal estimate of lost earnings or a calibrated equilibrium model.

Financing and ownership exposures are retained in the graph but do not automatically transmit the factory outage through identical equations. That restraint is essential. Common shareholders may experience a portfolio loss after prices respond, while lenders could incur credit risks if borrowers become impaired; neither mechanism equals the shortage of a physical component. Later analysis can layer the channels deliberately. For now we compute operational contagion first, then annotate potential financial amplifiers separately. An interpretable risk score is more valuable pedagogically than a composite number that silently mixes incompatible mechanisms.

The scenario should be read as a transparent chain of conditional assumptions. A seed value of 0.90 says how severely the selected fictional manufacturers are impaired at the start; it is not an estimate of the probability or scale of a real interruption. The edge weight represents the synthetic dependence assigned to one supplier–customer link, while the decay factor reduces downstream transmission at every hop. The hop cap and minimum threshold keep the exercise bounded and help students distinguish near exposures from remote, weak paths. Changing any one of these settings can change which firms appear in the ranking, so every score should be read together with its path and parameter choices.

The maximum-path rule answers a specific question: what is the strongest modeled route by which this firm could be exposed? It does not answer how much exposure arrives across all suppliers, whether alternative components are available, or how the firm would respond. The separate leverage and cash-buffer adjustment adds a simple vulnerability modifier, not a calibrated balance-sheet model. Students can compare the operational component with the final indicator to see how the proxy changes ordering, but they should not translate either value into expected loss, default risk, or a recommended position size. In later work, calibration would require verified input shares, inventories, substitution possibilities, contractual terms, and observed disruption outcomes.

Propagation mechanics and interpretation

Four semiconductor firms are chosen from the East Asian synthetic subset where available; the seed exposure is 0.90. A breadth-first queue follows supplier-to-customer edges. For each eligible downstream edge with synthetic weight w_{uv}, transmitted exposure is $x_v^{\text{candidate}} = x_u \, w_{uv} \, 0.62$. The procedure stops after three hops, rejects cycles within the current path and discards transmissions below 0.025. The recorded operational exposure is the *maximum* pathway contribution observed for each company, not the sum across all paths. Each path stores the ordered companies and supporting edge IDs.

The financial risk indicator then modifies operational exposure for leverage and cash buffers: $R_i = X_i(1+0.35L_i)/(1+0.30C_i)$. This is a teaching score, not a default probability, a predicted return or an estimated loss. The graph generates a traceable scenario hypothesis: it does not establish empirical causality.

Why the maximum matters

A firm may be reached through several supply routes, but this implementation retains its highest propagated exposure. Students should not read the resulting score as total aggregated network loss.

Cell 06 — Event-only and broad-document retrieval baselines

The sixth stage asks what an ordinary information system could deliver without traversing relational dependencies. Our first baseline contains only the event description and a generic risk question. That baseline offers almost no firm-specific evidence and is useful mainly as a reminder that convincing language is not the same as grounded analysis. The second baseline includes a broad sample of company notes. It supplies more corporate context, but the analyst must still discover which records actually connect to the disruption.

The purpose is not to make the simpler baselines intentionally poor. Long-context systems can work well when the relevant source material is included and the model attends properly to it. In this synthetic exercise, however, a finite context budget forces document selection. We create a deterministic broad sample and calculate how many of the independently defined high-exposure firms it includes. This allows students to observe how raw document volume can be less informative than purposeful evidence selection.

Both contexts are packaged in readable text files. They use the same event wording and do not assert findings that have not been computed. This last requirement matters: if a baseline prompt contains the benchmark conclusions, it no longer tests retrieval or reasoning. The exercise is designed to show the construction of contexts before the model receives them. This architectural separation allows later evaluation to distinguish failures of evidence discovery from failures of human or machine judgment.

These baselines are useful because they expose two common failure modes in knowledge work. With event-only context, a model may produce fluent generalities but have no basis for identifying particular downstream firms. With broad context, the evidence may technically be present yet remain difficult to locate within a fixed prompt. The latter resembles a crowded briefing pack: more pages do not guarantee that the decision-relevant record is surfaced. A deterministic sample makes this problem measurable, while keeping the contents independent of the later graph-ranked candidate list.

The 12,000-character limit is part of the experiment, not a universal threshold for language models. It creates a shared practical constraint under which the number and detail of dossiers compete for space. Registry order is intentionally simple and transparent, but it is not an optimal selection policy. A different order or budget could produce another baseline score. That sensitivity is worth teaching: retrieval performance depends not only on the documents that exist, but also on selection rules, truncation, prompt construction, and how the context is presented to a reader or model.

Learners should compare the methods without treating the baseline as a contest designed to make graph retrieval win. Broad-document context can be effective when an analyst already knows which records to include or when a corpus is short. Event-only context is valuable as a floor because it shows how little firm-specific reasoning is possible from an event description alone. Keeping these conditions separate lets the notebook ask whether structured evidence selection adds useful information, and later cells can compare that contribution against lexical and graph-based retrieval under the same synthetic event.

The question becomes one of evidence selection

The notebook holds the analytical question fixed while varying what evidence enters context. The event-only baseline supplies the synthetic disruption and its directly affected manufacturers. The broad-document baseline appends Markdown files in registry order until the context budget of 12,000 characters is reached. It does not prioritize company connectivity or stress rankings. Longer dossiers may therefore reduce the number of companies included, even as they increase background detail. The experiment separates the *availability of information* from the *relevance of information*.

What this cell does not yet do

Cell 06 prepares two user-message contexts. It does not calculate new propagation routes or invoke Claude. Model judgment enters at Cell 09, after the graph-aware package has been assembled.

Cell 07 — TF–IDF document retrieval

The seventh stage introduces a transparent document-retrieval baseline. TF–IDF represents documents and questions in a shared lexical space, giving higher weight to terms that are informative relative to the corpus. It is intentionally simpler than a modern dense embedding system, but that simplicity makes its strengths and limits easy to inspect. We combine each company's name, sector, region and description into its text representation. The query names the semiconductor interruption and a selection of associated industry terms.

A semantic failure occurs when a downstream firm is economically exposed through a relationship but its note does not use the same vocabulary as the event. The TF–IDF ranking may retrieve obvious manufacturers while overlooking logistics firms or financiers connected by several economically meaningful steps. This is not a failure of mathematics; it reflects the fact that the baseline has not been provided with structured relational knowledge. The graph-based strategy in the next cell will bring a different source of relevance into the evidence selection process.

We preserve the ranked identifiers and similarity scores, which allows the retrieval procedure to be audited independently from any model-generated language. We also avoid declaring high similarity equivalent to high financial exposure. A document can be linguistically relevant yet economically unimportant, and vice versa. Evaluating these two concepts separately is one of the core pedagogical objectives of the laboratory.

TF–IDF offers a useful middle ground between manual browsing and opaque semantic search. Term frequency records how often a word or phrase appears in a document, while inverse document frequency gives relatively greater weight to terms that are uncommon across the document collection. Combining these values produces vectors that can be compared with the event query using cosine similarity. The notebook's unigrams and bigrams allow it to match both individual terms and short phrases, but the ranking remains a measure of lexical resemblance. It does not encode supplier direction, input criticality, or the timing of a commercial relationship.

The top 25 candidates form a deliberately bounded retrieval set. Including a company name, sector, region, and generic description in each representation gives the query several transparent sources of matching terms. Supplying only the first 350 characters of each selected note limits context growth and makes truncation visible. An analyst can inspect the query, ranked IDs, scores, and selected snippets to understand why a candidate appeared. This kind of auditability is a benefit even when the result is incomplete: it makes retrieval errors discussable rather than disguising them behind a fluent summary.

Vocabulary mismatch is the central limitation. A logistics provider might be operationally important because it serves a chip-dependent manufacturer, yet its generic description may contain none of the query's semiconductor terms. Conversely, a company can discuss chips prominently without depending on any of the disrupted suppliers. Changes in wording, abbreviations, or language can also move a document up or down the ranking without any change in its underlying economics. TF–IDF is therefore a credible and interpretable baseline, but its scores should not be described as probabilities, risk magnitudes, or evidence that a firm is connected to a shock.

Lexical relevance is not economic exposure

TF–IDF is applied to name, sector, region and generic company description, using unigrams and bigrams. A query about semiconductors, production disruptions, components, automotive and East Asia is compared against each company representation using cosine similarity. The method ranks 25 candidates; the constructed context supplies their scores and only the first 350 characters of each company note. This economical baseline is transparent, repeatable and auditable. Yet it scores shared vocabulary, not whether the company is downstream of a disrupted supplier. Graph traversal introduces the missing relational criterion.

A useful counterexample

A geographically distant automotive company may not contain many semiconductor terms in its generic description while still receiving critical components along a two-hop supply path. Text matching can miss that link; a typed directed graph can represent it explicitly.

Cell 08 — Graph-aware evidence and traceable pathways

The eighth stage implements the intellectual center of the laboratory. The retrieval question is no longer simply "which documents contain terms similar to the event?" It becomes "which firms are connected to the disrupted manufacturers through plausible economic transmission paths, and what evidence supports each proposed connection?" The cell ranks observed synthetic exposures, retrieves the corresponding path ledger and constructs a compact package that a human or LLM can inspect.

There are two important constraints. First, traversal remains bounded by hop count and minimum transmitted intensity. Without those rules, a sufficiently dense business network would eventually make nearly every firm seem connected. Second, a pathway must expose its source identifiers and edge sequence. A conclusion like "Company 212 is affected through a second-tier component supplier" is not auditable unless the analyst can verify the alleged chain. Our context therefore presents the path and relation records alongside the risk indicators.

The graph-aware package is not independent from the propagation model; it deliberately uses its deterministic outputs. Consequently, comparing its exposure recall to document-only retrieval measures the information provided by this architecture, not a randomized superiority test. A stronger empirical study would withhold true outcomes, use alternative propagation models, vary graph quality and test external validity. Here we focus on a simpler and very valuable lesson: a governed Second Brain can assemble explicit, connected evidence for a new event that would be hard to reconstruct from an unordered pile of notes.

This cell changes the unit of retrieval from a document that resembles the query to a path that satisfies an explicit economic rule. The graph traversal is useful because the endpoint dossier need not repeat the event vocabulary: a downstream electronics assembler or distributor can be surfaced through its recorded route from a seed manufacturer. The explanatory value comes from showing not only the selected firm but the intermediate companies and source edges that make it relevant. Reviewers can then challenge any step in the route instead of accepting a bare ranked name.

The package is intentionally compact. Twenty five non-shocked firms are retained for ranking, while detailed context is prepared for the top fifteen; each selected firm can also carry up to three alternative high-intensity paths in the JSON evidence export. This separates a screening list from a more detailed review set. A risk team could use the shorter list to decide where to investigate and the path records to guide source verification. The numbers are notebook design choices, not thresholds that should be carried into other organizations without testing.

Graph-aware retrieval still inherits the assumptions and errors of the graph and propagation model. A missing edge can hide a true synthetic route; a false edge can create a spurious one; an inappropriate coefficient can distort rankings. Since the same model informs both selection and the benchmark, good agreement is partly built into the experiment. The proper conclusion is that typed paths can organize evidence in a useful and inspectable way under this scenario, not that the method has been proven superior on independent outcomes. Real deployment would need external source validation, dated updates, and blinded evaluation.

From relevant documents to relevant transmission paths

The graph-aware approach first excludes the directly shocked manufacturers from downstream discovery. It takes the 25 highest-ranked non-shocked firms under the simulated risk indicator, then packages detailed best-path information for the top 15. A supplied path includes the company ID, sector, operational exposure, stress indicator, ordered route back to an initial shock and corresponding source edge IDs. For each selected company, the separate JSON evidence export preserves up to three high-intensity paths. This is a compact record of *why* a firm entered the risk review.

Methodological caution

The graph-aware selector and the evaluation benchmark both use the same stress model. Higher synthetic recall therefore tests retrieval alignment with the model, not independently verified superiority or predictive performance in live markets.

Cell 09 — Bounded LLM judgment and market-risk memorandum

Only after evidence preparation do we ask the language model for judgment. Claude Haiku 5.5 is assigned the role of a cautious market-risk analyst. Its instructions require it to work exclusively with the synthetic event and supplied graph context, preserve firm and edge identifiers, distinguish operational exposure from price forecasts and suggest reversible, governed actions rather than executable trades. A successful response is a decision memorandum that can be read critically, not a claim of autonomous insight.

The system prompt requests six clearly distinguished sections: decision summary, exposure channels, evidence IDs, portfolio risk controls, uncertainty and human review. This format is intentionally parallel to the disciplined memorandum structures in Laboratory 01. The prompt also requires acknowledgment when liquidity, ownership or real-time prices are missing. That prevents a rich-looking dataset from being mistaken for comprehensive investment intelligence.

The notebook reads `ANTHROPIC_API_KEY` from Colab Secrets and uses the Messages API. It avoids passing unsupported temperature controls and does not depend on provider-enforced JSON output. The cell handles multiple possible text blocks, records the response or the error, and never fabricates a model judgment if the API is unavailable. The complete deterministic experiment still works. A later course can add human rating of the memo, more realistic portfolio constraints and model comparisons, but the present objective is to demonstrate how external knowledge becomes bounded, inspectable language-model reasoning.

The prompt acts as a boundary around the model's authority. It gives the system a defined analytical role, a fixed event, and a selected evidence package, then asks for a memorandum that distinguishes facts supplied by the notebook from interpretation. Requiring firm IDs and edge IDs helps preserve traceability when language is compressed. Requiring uncertainty and missing information helps prevent the polished form of a memo from implying that the dataset is complete. These instructions do not guarantee perfect compliance, which is why the generated text remains a draft for review rather than an authoritative determination.

Optional access is an operational design choice as well as a privacy control. The deterministic notebook should still run when there is no API key, when the provider is unreachable, or when the response cannot be parsed as expected. Recording status and errors allows students to distinguish a model failure from an empty risk result. The notebook's handling of multiple text blocks is a practical reminder that service responses have structure; consuming them safely requires checking the returned content instead of assuming every request yields one clean string.

Human review should focus on both factual fidelity and reasoning quality. Reviewers can verify that cited paths exist, that the model has not upgraded a synthetic indicator into a forecast, and that proposed safeguards are reversible and within the institution's authority. They should also ask what the model omitted, whether it confused operational and financial channels, and whether uncertainty is proportionate to the evidence. The exercise does not claim that a cautious prompt eliminates hallucinations or bias. Its contribution is to make the model's limited role, inputs, and failure state visible enough for a reviewer to challenge.

What is added by bounded model judgment

The LLM is neither the source of the network facts nor the propagation calculator. It receives the previously prepared `D_graph_aware` context and a system instruction to use supplied evidence only. It must distinguish synthetic exposure indicators from probabilities, market forecasts and real trading advice. The requested memorandum covers decision summary, transmission paths and edge IDs, portfolio controls, missing information and human review. The API call may be absent or fail; in that case the notebook saves its status and does not pretend a memorandum was generated.

Intelligence meets knowledge

Knowledge: entity records, documented relationship IDs and graph-derived pathways. **Analysis:** bounded propagation and ranking. **Judgment:** synthesizing those records into defensible questions, controls and uncertainty statements. **Governance:** a human validates assumptions before acting.

Cell 10 — Evaluate retrieval and export the laboratory

The final cell turns a demonstration into an auditable experiment. It defines a synthetic relevance benchmark from the firms most exposed under the propagation model, excluding the initial disrupted manufacturers. For each context method it calculates retrieval recall, evidence density and document-selection size. These are architectural diagnostic metrics; because the ground truth derives from the same simulated network that feeds graph retrieval, they are not independent estimates of investment performance.

The evaluation also records whether a Claude memorandum was actually returned. A model error is not replaced with a predetermined answer or a misleading success flag. The export includes the complete Obsidian vault, company registry, edge ledger, event record, exposure scores, ranking tables, evidence paths, model status and all context packages. This zip file allows a student to inspect the assumptions outside the notebook and to repeat experiments with modified source data.

The laboratory closes with a deliberately practical counterfactual agenda. Remove a high-confidence supply edge; alter the shock location; reduce the propagation decay; or test a company whose historical covariance is modest but whose dependency path is short. Ask whether the recommended risk review changes for good evidentiary reasons. That practice is more important than celebrating the largest-looking retrieval score. A well-designed Second Brain is useful because it exposes assumptions, invites correction and supports accountable decisions under uncertainty.

The evaluation's benchmark is operationally convenient because it can be computed directly from the synthetic stress scores, but that convenience creates dependence between the test and one retrieval strategy. Graph-aware retrieval is designed around the same modeled exposure pathways used to label the critical set, so high overlap is evidence of consistency with the chosen model. It is not independent confirmation that the graph is correct, that the synthetic coefficients are realistic, or that the selected firms would experience losses in a live shock. An honest report should state this limitation alongside any comparison rather than presenting one metric as a performance claim.

The evidence-density measure adds a context-efficiency perspective by counting matched critical sources relative to the amount of text supplied. It can help distinguish a compact package that surfaces relevant records from a large package that happens to contain them somewhere. Yet density also depends on what counts as a source, how context length is measured, and which benchmark labels are used. Document count, overlap, and characters should therefore be reported together; none alone captures the quality of the evidence, the truth of a path, or the usefulness of a human decision.

The exported archive supports a more meaningful next experiment than rerunning a cell and looking only at a new score. Students can remove or alter one edge, change the event seeds, vary the decay or threshold, and then compare the registry, paths, contexts, metrics, and resulting memo status. If the result changes, the archive makes it possible to locate which assumption changed and which artifacts depend on it. That is the core of reproducible challenge: preserving enough intermediate evidence that another analyst can reconstruct, question, and improve the analysis without trusting a final narrative alone.

What the metrics actually establish

The benchmark comprises the top 20 non-shocked firms under the laboratory's synthetic risk indicator. For each of four strategies, the notebook reports the number of selected documents, their overlap with those 20 firms, the total context size and matched critical sources per thousand characters. These are retrieval diagnostics; they do not grade Claude's memo, measure out-of-sample forecasts or establish that graph-based risk assessments are correct in actual markets. The final ZIP includes the vault, registry, ledger, event, exposure rankings, contexts, evidence paths, model status and README so that an analyst can challenge the scenario and repeat the work.

Next experiment

To examine the *quality of judgment*, submit comparable evidence packages to the same model under controlled prompts and assess the resulting memos against a separately defined rubric for fidelity, completeness, uncertainty and proposed safeguards.

Conclusion: From connected knowledge to governed market judgment

The second laboratory has transformed a synthetic universe of five hundred companies into a deliberately inspectable institutional knowledge system. Its central artifact is not the risk ranking on its own, nor the optional memorandum written by Claude. The core achievement is a chain of evidence: company record, typed relationship, operational scenario, propagation calculation, retrieval package, model judgment, evaluation and export. Each stage can be questioned separately. This is what makes the Second Brain conception useful to a professional trading desk or risk committee: intelligence is generated through an accountable knowledge architecture rather than by asking a conversational model to improvise a compelling story.

The market setting highlights a limitation of exclusively price-based analysis. Correlation and covariance summarize associations observed over historical windows. They are valuable but do not necessarily disclose the exact dependencies through which a novel event could spread. Supplier–customer contracts, input concentration, geographic exposure, lending relationships and ownership structures may provide additional information about future vulnerabilities. Our synthetic semiconductor shock illustrates how a graph can surface indirect exposures across multiple tiers even when the affected firms are not immediate lexical matches for the crisis narrative. It does not establish that these pathways will predict future prices, and the notebook is careful not to say otherwise.

A methodological achievement is the distinction between economic mechanisms. The network includes supply, financing and ownership links, but its operational stress model propagates only along the supply layer. This choice is conservative and pedagogically important. A physical input disruption, a bank credit event and a common-shareholder selloff transmit risk in different ways, over different horizons and with different empirical parameters. Combining them without calibration would make the algorithm look more comprehensive while making it less interpretable. An advanced laboratory could estimate a multiplex contagion process; this introductory experiment instead prioritizes causal discipline and transparent assumptions.

Retrieval is also more nuanced than a contest between fashionable technologies. The event-only condition provides little firm evidence. Broad-document assembly may contain relevant notes without making their pathways prominent. TF–IDF can identify obvious sector matches and offers a simple, defensible baseline. The graph-aware method makes a different move: it retrieves connections supported by typed relationships and packages the path identifiers, weighted exposure and firm descriptions together. Its apparent recall advantage under the synthetic benchmark should be read as evidence of architectural fit, not as a validated performance claim. An honest analyst would demand independent, time-stamped outcomes before claiming an investment edge.

Claude Haiku 5.5 enters only after this deterministic infrastructure is built. Its responsibility is to translate a bounded evidence package into a compact risk memorandum, while stating missing information and recommending reviewable controls. The model is not granted broker access, execution authority or permission to invent price histories. If the API fails or an authorized key is absent, the notebook records the failure and proceeds with its auditable diagnostics. This operational separation is a cornerstone of trustworthy AI in finance: retrieval, computation, judgment and action need not share a single uncontrolled pathway.

Institutional knowledge brings a further advantage: it can outlive the immediate analyst and the immediate shock. A stored supplier dependency may become relevant again during an unrelated geopolitical event. A historic decision memorandum may explain why an investment committee rejected a concentration risk. A provenance record may reveal that a presumed supplier relationship was never verified. In a real firm, these insights would require governance structures: data stewards, source certification, effective-date management, permissions, periodic review and explicit human responsibility for conclusions. Without those controls, a persistent Second Brain may become a persistent archive of mistakes.

Students should finish by changing one assumption and rerunning the entire exercise. Compare a one-hop outage with a three-hop disturbance; replace the seed manufacturers; remove a critical supplier; or reduce the confidence attached to a disputed link. Observe whether the retrieval package changes, whether Claude's memorandum acknowledges the change, and whether the exported evidence provides enough information to explain the difference. Those experiments reveal more about intelligence than a polished static answer does.

The progression from Laboratory 01 to Laboratory 02 is therefore conceptual as well as technical. Regulatory intelligence required reconstructing authority, constraints and permissible action. Market intelligence requires reconstructing economic dependencies, transmission mechanisms and bounded risk responses. In both cases, the same principle holds: **knowledge becomes operationally useful when its relationships are explicit, its evidence is traceable, and its interpretation is governed.** The Second Brain does not replace expertise; it equips expertise with a persistent structure that can be explored, corrected and reused.

Selected readings

1. Allen, F., and Gale, D. (2000). Financial contagion. *Journal of Political Economy*, 108(1), 1–33.
2. Acemoglu, D., Ozdaglar, A., and Tahbaz-Salehi, A. (2015). Systemic risk and stability in financial networks. *American Economic Review*, 105(2), 564–608.
3. Carvalho, V. M. (2014). From micro to macro via production networks. *Journal of Economic Perspectives*, 28(4), 23–48.
4. Elliott, M., Golub, B., and Jackson, M. O. (2014). Financial networks and contagion. *American Economic Review*, 104(10), 3115–3153.
5. Battiston, S., et al. (2012). DebtRank: Too central to fail? *Scientific Reports*, 2, 541.
6. Page, L., Brin, S., Motwani, R., and Winograd, T. (1999). The PageRank citation ranking: Bringing order to the Web. Stanford Technical Report.
7. Newman, M. E. J. (2018). *Networks* (2nd ed.). Oxford University Press.
8. Lewis, P., et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *NeurIPS*.
9. Robertson, S. (2004). Understanding inverse document frequency. *Journal of Documentation*, 60(5), 503–520.
10. Pearl, J. (2009). *Causality* (2nd ed.). Cambridge University Press.
11. Brunnermeier, M. K., and Pedersen, L. H. (2009). Market liquidity and funding liquidity. *Review of Financial Studies*, 22(6), 2201–2238.
12. Jackson, M. O. (2008). *Social and Economic Networks*. Princeton University Press.
13. Basel Committee on Banking Supervision (2013). *Principles for effective risk data aggregation and risk reporting* (BCBS 239).
14. National Institute of Standards and Technology (2023). *AI Risk Management Framework (AI RMF 1.0)*.
15. Obsidian (n.d.). *Help: Internal links and properties*. Documentation.

THE FIVE-LABORATORY PEDAGOGICAL TRAJECTORY

This is Laboratory 02 within the Second Brain collection. It demonstrates an economic case in which the institution must relate many entities, distinguish multiple evidence channels, and reconstruct the provenance of a risk hypothesis. The generalizable method is a sequence of governed knowledge operations: *vault* → *typed graph* → *scenario* → *retrieval* → *bounded judgment* → *audit*. The precise contents of other laboratories are not presumed by this monograph.

Source and methodological reference

1. Alejandro Reynoso, *The Trading Desk That Sees Contagion Before Prices Do: Market Intelligence through an Institutional Second Brain*, companion Google Colab notebook `TRADING_CONTAGION_SECOND_BRAIN.ipynb`, October 2026. Primary source for every cell narrative, operational parameter, output file and experiment discussed in this monograph.

2. Supplied figures, `cell_1.png`–`cell_10.png`, one figure mapped to each laboratory stage. These are pedagogical illustrations, not empirical results.

3. Reference visual style: *Self-Driving Delivery Network: Optimization with Judgment*, supplied LaTeX monograph; navy, teal, restrained rules, boxed concepts, airy spacing and document hierarchy have been adapted here.